

CO-1 AT-3

QUESTION -1

CODE:

```
# Load Library
library(ggplot2)

# Dataset
retail <- data.frame(
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
  Region=c("North","South","East","West","North","South","East","West",
"North","South","East","West","North","South","East","West",
"North","South","East","West"),
  Category=c("Electronics","Furniture","Clothing","Food","Electronics",
"Furniture","Clothing","Food","Electronics","Furniture",
"Clothing","Food","Electronics","Furniture","Clothing",
"Food","Electronics","Furniture","Clothing","Food"),
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
560000,670000,195000,980000,430000,760000,280000,1150000,
510000,690000,220000,1250000),
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
72000,98000,-8000,205000,51000,132000,24000,250000,
69000,115000,15000,285000),
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
2200,3900,1600,6100)
)

# Retail Dashboard Visualization
ggplot(retail,
```

```

aes(x = Sales,
    y = Profit,
    size = Customers,
    color = Region,
    shape = Category)) +
geom_point(alpha = 0.9) +
geom_text(aes(label = Branch),
    vjust = -1,
    size = 3.5,
    color = "black") +
scale_size(range = c(4, 14)) +
scale_color_manual(values = c(
    "North" = "#1E88E5",
    "South" = "#E91E63",
    "East" = "#43A047",
    "West" = "#FB8C00"
)) +
scale_shape_manual(values = c(
    "Electronics" = 16,
    "Furniture" = 17,
    "Clothing" = 15,
    "Food" = 18
)) +
scale_x_log10() +
labs(
    title = "Retail Sales Performance Dashboard",
    subtitle = "Sales vs Profit with Customers, Region and Category",
    x = "Sales (₹) - Log Scale",
    y = "Profit (₹)",

```

```
color = "Region",
shape = "Category",
size = "Customers"
)
theme_minimal(base_size = 15) +
theme(
  plot.title = element_text(size = 20, face = "bold", color = "darkblue"),
  plot.subtitle = element_text(size = 13, color = "darkred"),
  axis.title = element_text(face = "bold"),
  legend.position = "right",
  panel.grid.major = element_line(color = "gray85"),
  panel.grid.minor = element_blank(),
  plot.background = element_rect(fill = "#F8F9FA", color = NA)
```

EXPLANATION:

1. library(ggplot2)

- Loads the ggplot2 package.
 - Used to create attractive and professional visualizations.
-

2. retail <- data.frame(...)

- Creates a data frame named retail.
 - Stores Branch, Region, Category, Sales, Profit, and Customers data.
-

3. ggplot(retail, aes(...))

- Initializes the graph.
- Maps variables to visual aesthetics.
- Sales → X-axis.
- Profit → Y-axis.
- Customers → Bubble size.
- Region → Bubble color.

- Category → Bubble shape.
-

4. `geom_point(alpha=0.9)`

- Draws the bubbles (points).
 - `alpha=0.9` adds slight transparency.
 - Makes overlapping bubbles easier to see.
-

5. `geom_text(aes(label=Branch))`

- Displays branch names (B1, B2, B3, etc.).
 - Places labels above each bubble for easy identification.
-

6. `scale_size(range=c(4,14))`

- Controls the minimum and maximum bubble size.
 - Larger bubbles represent more customers.
-

7. `scale_color_manual()`

- Assigns different colors to each region.
 - Helps distinguish North, South, East, and West easily.
-

8. `scale_shape_manual()`

- Assigns different shapes for each product category.
 - Makes Electronics, Furniture, Clothing, and Food easy to identify.
-

9. `scale_x_log10()`

- Applies a logarithmic scale to the Sales axis.
 - Compresses large sales values.
 - Improves comparison between small and large branches.
-

10. `labs()`

- Adds a graph title.
 - Adds a subtitle.
 - Labels the X-axis and Y-axis.
 - Displays legend titles.
-

11. theme_minimal()

- Applies a clean and professional theme.
 - Removes unnecessary background elements.
 - Improves graph readability.
-

Aesthetic Mapping

- X-axis → Sales
 - Y-axis → Profit
 - Bubble Size → Customers
 - Bubble Color → Region
 - Bubble Shape → Category
 - Branch Labels → Branch Names
-

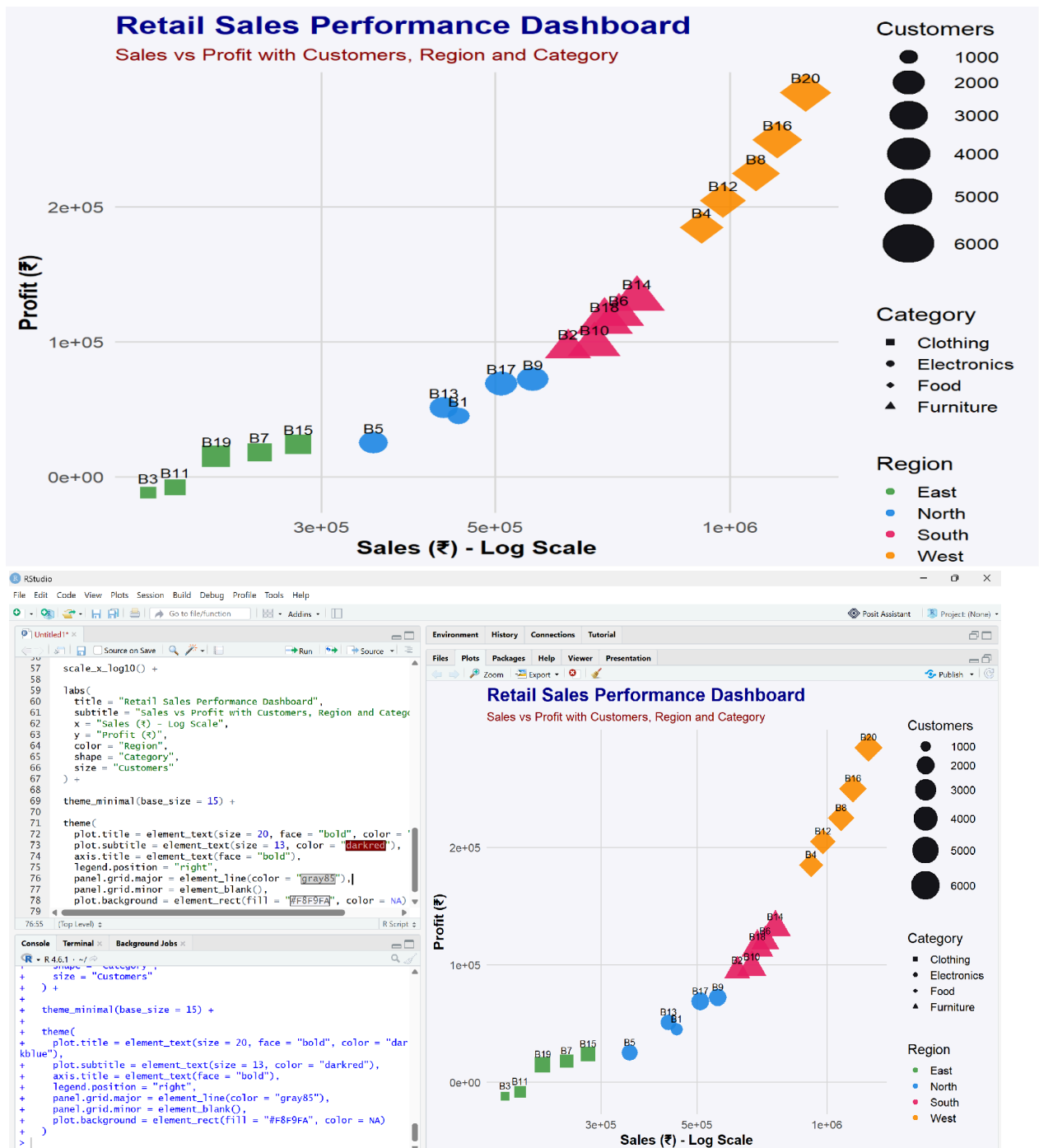
Why Logarithmic Scale?

- Sales values vary widely.
 - Compresses large values.
 - Makes all branches visible.
 - Improves comparison.
-

Advantages of the Visualization

- Displays five variables in a single chart.
- Easy to compare sales and profit.
- Bubble size shows customer count.
- Colors identify regions.

- Shapes distinguish product categories.
- Professional and easy to interpret.



QUESTION 2

CODE:

```
# Load Library

library(ggplot2)

# Dataset

energy <- data.frame(

  Month = month.abb,

  Coal = c(620,580,590,605,640,670,690,720,700,680,650,630),

  Gas = c(310,295,305,320,340,355,360,370,365,350,335,320),

  Solar = c(45,60,90,130,180,230,250,245,210,160,90,50),

  Wind = c(80,95,120,150,170,185,195,190,175,150,110,90),

  Hydro = c(140,150,160,170,180,195,205,210,200,185,165,150)

)

# Convert data into long format

energy_long <- data.frame(

  Month = rep(energy$Month, 5),

  Source = rep(c("Coal", "Gas", "Solar", "Wind", "Hydro"),

    each = nrow(energy)),

  Generation = c(

    energy$Coal,

    energy$Gas,

    energy$Solar,

    energy$Wind,

    energy$Hydro

  )

)

#####

# Q1 Cartesian Coordinate Visualization
```

```
#####
```

```
ggplot(energy_long,  
  aes(x = Month,  
    y = Generation,  
    group = Source,  
    color = Source)) +  
geom_line(size = 1.5) +  
geom_point(size = 3) +  
scale_color_manual(values = c(  
  "Coal" = "#4E342E",  
  "Gas" = "#1E88E5",  
  "Solar" = "#FDD835",  
  "Wind" = "#00ACC1",  
  "Hydro" = "#43A047"  
)) +  
labs(  
  title = "Monthly Electricity Generation",  
  subtitle = "Cartesian Coordinate System",  
  x = "Month",  
  y = "Generation (MW)",  
  color = "Energy Source"  
) +  
theme_minimal(base_size = 15) +  
theme(  
  plot.title = element_text(face = "bold",  
    size = 20,  
    color = "darkblue"),  
  plot.subtitle = element_text(size = 13),  
  axis.title = element_text(face = "bold"),
```



```

    legend.position = "right"
  )
#####

# Q2 Polar Coordinate Visualization
#####

ggplot(energy_long,
  aes(x = Month,
      y = Generation,
      fill = Source)) +
  geom_bar(stat = "identity",
    position = "stack",
    color = "white") +
  coord_polar() +
  scale_fill_manual(values = c(
    "Coal" = "#4E342E",
    "Gas" = "#1E88E5",
    "Solar" = "#FDD835",
    "Wind" = "#00ACC1",
    "Hydro" = "#43A047"
  )) +
  labs(
    title = "Electricity Generation by Energy Source",
    subtitle = "Polar Coordinate System",
    fill = "Energy Source"
  ) +
  theme_minimal(base_size = 15) +
  theme(
    plot.title = element_text(face = "bold",
      size = 20,

```

```

        color = "darkred"),
    plot.subtitle = element_text(size = 13),
    axis.title = element_blank(),
    axis.text = element_blank(),
    axis.ticks = element_blank()
)

#####

# Q3 Comparison

#####

cat("Comparison:\n")

cat("- Cartesian chart provides accurate comparison of monthly electricity generation.\n")

cat("- Polar chart provides an attractive circular representation but makes exact comparison
more difficult.\n\n")

#####

# Q4 Appropriate Coordinate System

#####

cat("Most Appropriate Coordinate System:\n")

cat("Cartesian Coordinate System\n\n")

#####

# Q5 Justification

#####

cat("Justification:\n")

cat("Cartesian coordinates allow easy comparison of energy generation across months and
energy sources using a common scale.\n")

cat("Polar coordinates are visually appealing but are better suited for cyclical or directional
data rather than precise quantitative comparison.\n")

```

EXPLANATION:

1. library(ggplot2)

- Loads the ggplot2 package.

- Used to create attractive and professional visualizations.
-

2. `energy <- data.frame(...)`

- Creates a data frame named `energy`.
 - Stores monthly electricity generation data for:
 - Coal
 - Gas
 - Solar
 - Wind
 - Hydro
-

3. `energy_long <- data.frame(...)`

- Converts the dataset into long format.
 - Makes it easier to plot multiple energy sources using `ggplot2`.
 - Each row contains:
 - Month
 - Energy Source
 - Generation Value
-

Cartesian Coordinate Visualization

4. `ggplot(energy_long, aes(...))`

- Initializes the graph.
 - Maps:
 - Month → X-axis
 - Generation → Y-axis
 - Energy Source → Color
-

5. `geom_line(size=1.5)`

- Draws a line for each energy source.

- Shows the monthly generation trend.
-

6. `geom_point(size=3)`

- Adds points to each month's value.
 - Makes individual observations easy to identify.
-

7. `scale_color_manual()`

- Assigns different colors to each energy source.

Energy Source Color

Coal	Brown
Gas	Blue
Solar	Yellow
Wind	Cyan
Hydro	Green

8. `labs()`

- Adds graph title.
 - Adds subtitle.
 - Labels X-axis and Y-axis.
 - Displays legend title.
-

9. `theme_minimal()`

- Applies a clean and professional appearance.
 - Improves readability.
-

Polar Coordinate Visualization

10. `geom_bar(stat="identity")`

- Creates stacked bars using the generation values.
- Each bar represents a month.

11. coord_polar()

- Converts the bar chart into a circular (polar) chart.
 - Displays data in a radial format.
-

12. scale_fill_manual()

- Assigns different colors to each energy source.
 - Improves distinction between categories.
-

13. theme_minimal()

- Gives the polar chart a clean appearance.
 - Removes unnecessary visual clutter.
-

Comparison of Both Visualizations

Cartesian Coordinate System

- Easy to compare monthly electricity generation.
 - Clearly shows increasing and decreasing trends.
 - Provides accurate quantitative comparison.
-

Polar Coordinate System

- Attractive circular representation.
 - Suitable for cyclical or seasonal data.
 - Exact value comparison is more difficult.
-

Which Coordinate System is More Appropriate?

- Cartesian Coordinate System
-

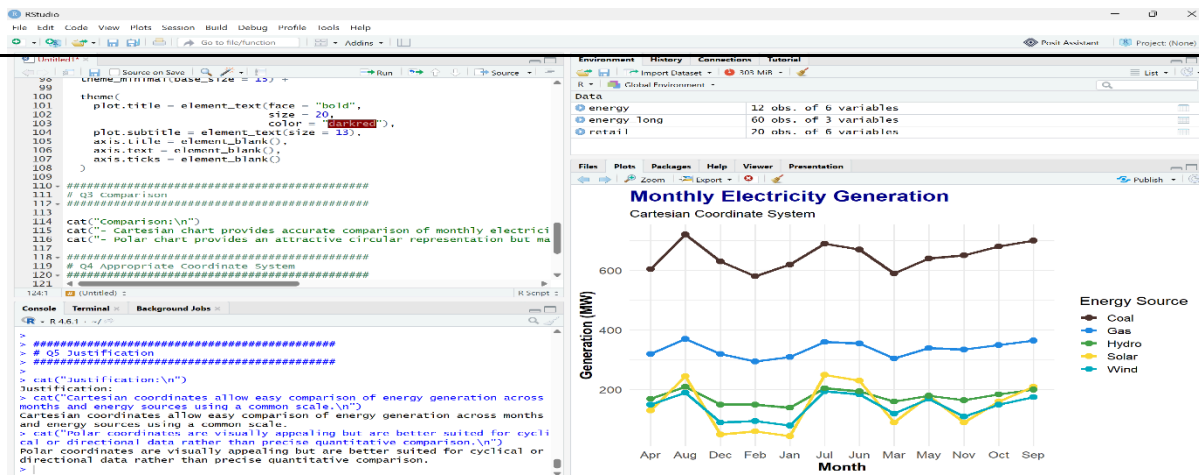
Justification

- Provides accurate comparison using a common scale.

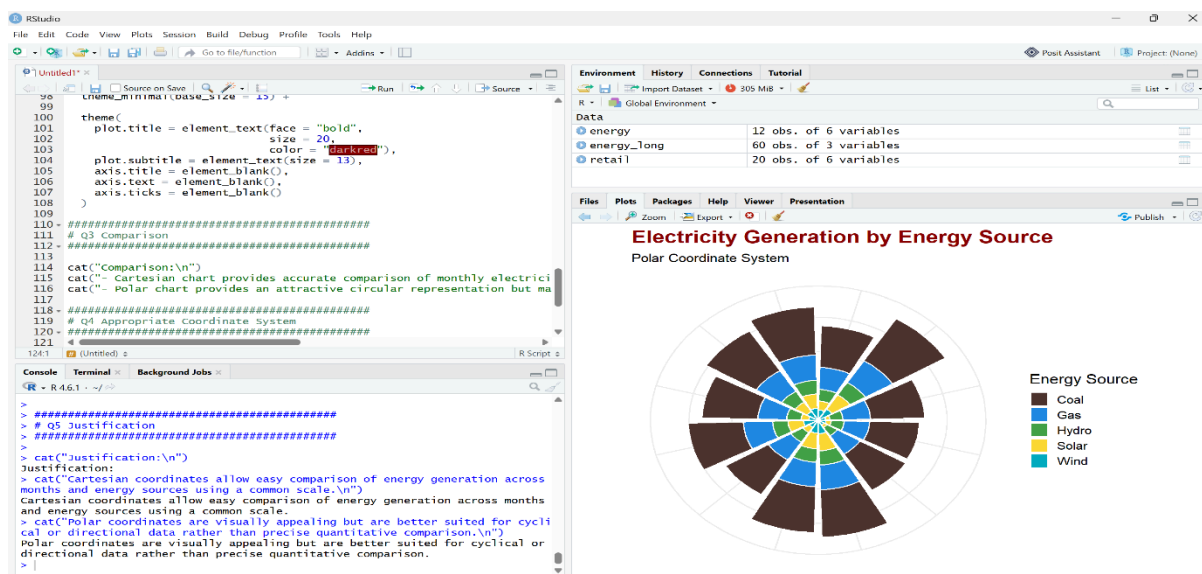
- Clearly shows trends across months.
- Easier to interpret numerical values.
- Better for analytical purposes.

Advantages of the Visualization

- Displays multiple energy sources together.
- Different colors distinguish each source.
- Line chart clearly shows trends over time.
- Polar chart provides an attractive circular view.
- Both visualizations complement each other.



QUESTION 3



CODE:

```

# Load Library

library(ggplot2)

# Dataset

climate <- data.frame(
  City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",
    "Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",
    "Bhopal","Patna","Goa","Kochi","Nagpur","Indore",
    "Surat","Visakhapatnam","Coimbatore","Mysuru"),
  Temperature=c(44,35,38,40,39,31,33,43,45,39,
    38,40,32,30,41,37,36,34,29,28),
  Humidity=c(22,84,78,72,48,66,58,26,18,42,
    36,45,89,91,30,34,76,82,68,64),
  Rainfall=c(6,280,220,190,85,120,95,2,1,35,
    48,72,360,410,12,20,180,240,130,145),
  AQI=c(365,110,125,180,160,72,84,290,320,240,
    175,220,58,52,270,190,145,98,60,55)
)

#####

# Climate Visualization

#####

ggplot(climate,
  aes(x=Temperature,
    y=AQI,
    size=Rainfall,
    color=Humidity))+
  geom_point(alpha=0.85)+
  geom_text(aes(label=City),
    vjust=-1,
    size=3.5,

```

```

        color="black")+
scale_size(range=c(4,15))+
scale_color_gradient(
low="#00B4D8",
high="#D00000"
)+
labs(
title="Climate Monitoring Dashboard",
subtitle="Temperature, AQI, Rainfall and Humidity",
x="Temperature (°C)",
y="Air Quality Index (AQI)",
size="Rainfall (mm)",
color="Humidity (%)"
)+
theme_minimal(base_size=15)+
theme(
plot.title=element_text(face="bold",
size=20,
color="darkblue"),
plot.subtitle=element_text(size=13,
color="darkred"),
axis.title=element_text(face="bold"),
legend.position="right",
panel.grid.major=element_line(color="grey80"),
panel.grid.minor=element_blank()
)
#####

# Question 3

#####

```



```

cat("Interpretation:\n")

cat("- Higher temperatures generally correspond to higher AQI in some cities.\n")

cat("- Larger bubbles indicate higher rainfall.\n")

cat("- Darker colors represent higher humidity.\n\n")

#####

# Question 4

#####

cat("Continuous Variables:\n")

cat("Temperature, Humidity, Rainfall, AQI\n\n")

cat("Categorical Variable:\n")

cat("City\n\n")

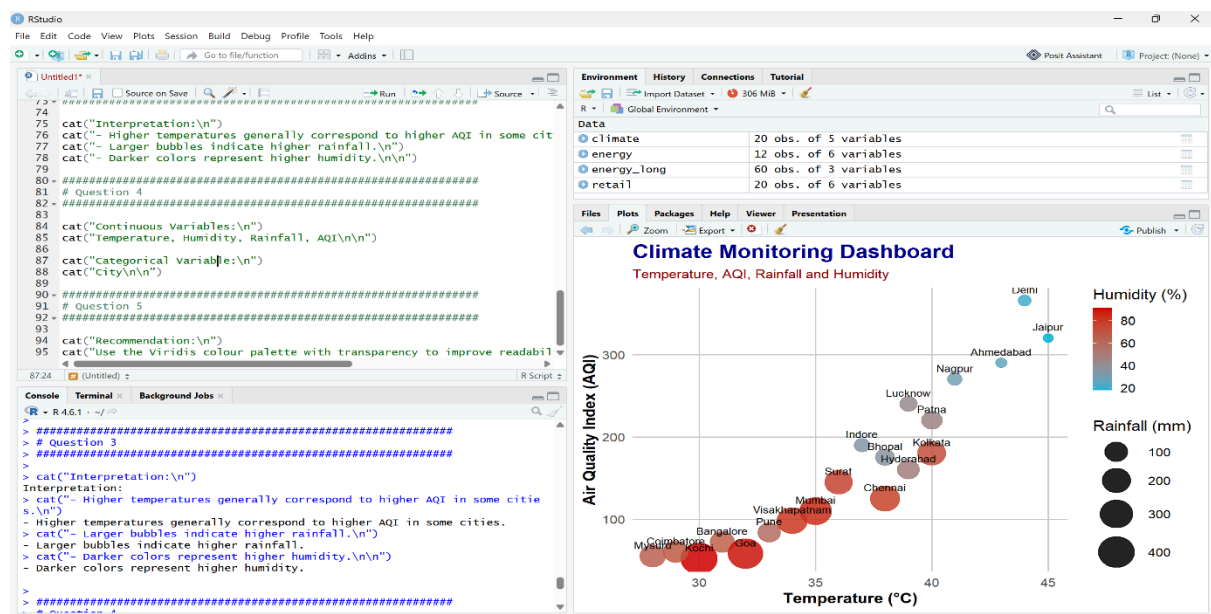
#####

# Question 5

#####

cat("Recommendation:\n")

```



EXPLANATION:

1. library(ggplot2)

- Loads the ggplot2 package.
- Used to create attractive and professional visualizations.

2. `climate <- data.frame(...)`

- Creates a data frame named `climate`.
- Stores data for:
 - City
 - Temperature
 - AQI (Air Quality Index)
 - Rainfall
 - Humidity

3. `ggplot(climate, aes(...))`

- Initializes the graph.
- Maps variables to visual aesthetics.
- Temperature → X-axis.
- AQI → Y-axis.
- Rainfall → Bubble size.
- Humidity → Bubble color.

4. `geom_point(alpha=0.85)`

- Draws the bubbles (points).
- `alpha=0.85` adds slight transparency.
- Makes overlapping bubbles easier to see.

5. `geom_text(aes(label=City))`

- Displays city names beside each bubble.
- Makes it easy to identify each city's data.

6. `scale_size(range=c(4,14))`

- Controls the minimum and maximum bubble size.

- Larger bubbles represent higher rainfall.
-

7. scale_color_gradient()

- Applies a color gradient based on humidity.
 - Lower humidity is shown with lighter colors.
 - Higher humidity is shown with darker colors.
-

8. labs()

- Adds the graph title.
 - Adds a subtitle.
 - Labels the X-axis and Y-axis.
 - Displays legend titles.
-

9. theme_minimal()

- Applies a clean and professional theme.
 - Improves readability by removing unnecessary background elements.
-

Aesthetic Mapping

- X-axis → Temperature (°C)
 - Y-axis → AQI
 - Bubble Size → Rainfall (mm)
 - Bubble Color → Humidity (%)
 - Labels → City Names
-

Interpretation

- Higher temperatures can be compared with AQI levels.
- Larger bubbles indicate cities with more rainfall.
- Darker colors represent higher humidity.
- City labels help identify individual observations.

Variables Used

Continuous Variables

- Temperature
- AQI
- Rainfall
- Humidity

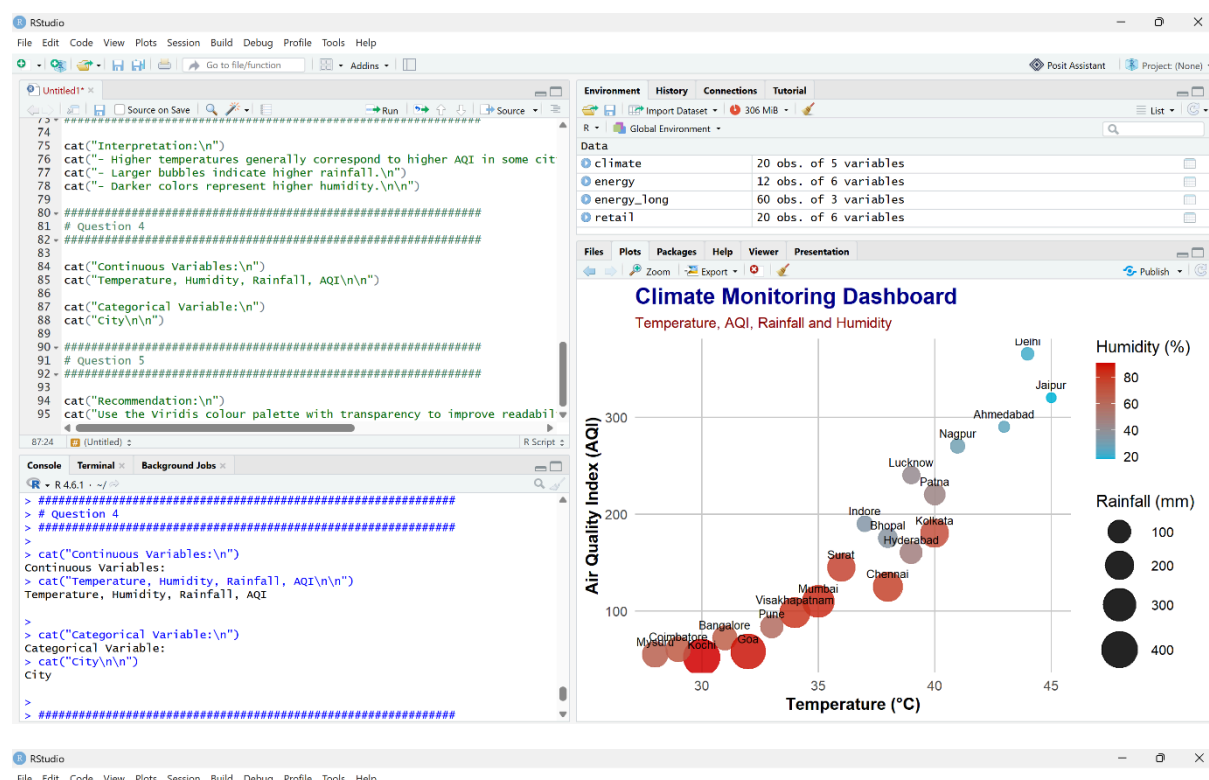
Categorical Variable

- City
-

Advantages of the Visualization

- Displays multiple climate variables in a single chart.
- Bubble size highlights rainfall differences.
- Color gradient shows humidity levels clearly.
- City labels improve identification.
- Makes comparison between cities easy.

cat("Use the Viridis colour palette with transparency to improve readability and accessibility for colour-blind users.\n")



QUESTION 4

CODE:

```
# Load Library
```

```
library(ggplot2)
```

```
# Dataset
```

```
software <- data.frame(
```

```
Team=paste("T",1:20,sep=""),
```

```
LinesOfCode=c(12000,18000,25000,35000,48000,
```

```
62000,81000,96000,120000,150000,
```

```
185000,240000,310000,390000,480000,
```

```
620000,810000,1050000,1350000,1800000),
```

```
Bugs=c(35,52,60,72,88,
```

```
110,125,142,165,188,
```

```
205,240,280,320,365,
```

```
420,490,560,640,720),
```

```
Developers=c(5,7,9,10,12,
```

```
14,16,18,20,22,
```

```
25,28,30,34,38,
```

```
42,46,50,55,60),
```

```
Complexity=c(2.1,2.4,2.7,3.0,3.2,  
3.5,3.8,4.1,4.5,4.8,  
5.1,5.4,5.8,6.2,6.6,  
7.1,7.6,8.0,8.5,9.0)  
)
```

```
#####  
# Software Analytics Visualization  
#####
```

```
ggplot(software,  
aes(x=LinesOfCode,  
y=Bugs,  
size=Developers,  
color=Complexity))+
```

```
geom_point(alpha=0.85)+
```

```
geom_text(aes(label=Team),  
vjust=-1,  
size=3.5,  
color="black")+
```

```
scale_x_log10()+
```

```
scale_size(range=c(4,15))+
```

```
scale_color_gradient(
```

```
low="#00C9A7",
```

```
high="#D90429"
```

```
)+
```

```
labs(
```

```
title="Software Development Analytics Dashboard",
```

```
subtitle="Logarithmic Scale for Lines of Code",
```

```
x="Lines of Code (Log Scale)",
```

```
y="Number of Bugs",
```

```
size="Developers",
```

```
color="Complexity"
```

```
)+
```

```
theme_minimal(base_size=15)+
```

```
theme(
```

```
plot.title=element_text(face="bold",
```

```
size=20,
```

```
color="darkblue"),
```

```
plot.subtitle=element_text(size=13,
```

```
color="darkred"),
```

```
axis.title=element_text(face="bold"),
```

```
legend.position="right",
```

```
panel.grid.major=element_line(color="grey80"),
```

```
panel.grid.minor=element_blank()
)
```

```
#####
```

```
# Question 3
```

```
#####
```

```
cat("Interpretation:\n")
```

```
cat("- Projects with higher Lines of Code generally contain more bugs.\n")
```

```
cat("- Larger bubbles represent teams with more developers.\n")
```

```
cat("- Darker colours indicate higher software complexity.\n\n")
```

```
#####
```

```
# Question 4
```

```
#####
```

```
cat("Why Logarithmic Scale?\n")
```

```
cat("- Lines of Code range from 12,000 to 1,800,000.\n")
```

```
cat("- Log transformation compresses large values.\n")
```

```
cat("- Makes comparison easier.\n")
```

```
cat("- Prevents smaller projects from appearing clustered together.\n\n")
```

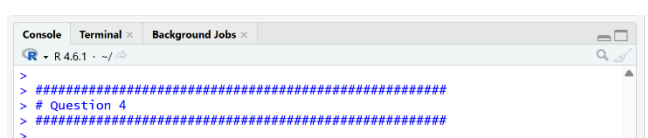
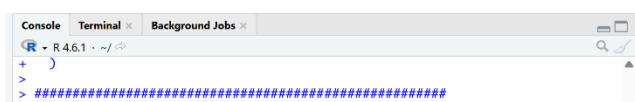
```
#####
```

```
# Question 5
```

```
#####
```

```
cat("Conclusion:\n")
```

```
cat("A logarithmic scale provides better readability and interpretation than a linear scale for highly skewed software project data.\n")
```



EXPLANATION:

1. `library(ggplot2)`

- Loads the ggplot2 package.
 - Used to create attractive and professional visualizations.
-

2. `software <- data.frame(...)`

- Creates a data frame named software.
 - Stores details of:
 - Team
 - Lines of Code (LOC)
 - Bugs
 - Developers
 - Complexity
-

3. `ggplot(software, aes(...))`

- Initializes the graph.
 - Maps variables to visual aesthetics.
 - Lines of Code → X-axis.
 - Bugs → Y-axis.
 - Developers → Bubble size.
 - Complexity → Bubble color.
-

4. `geom_point(alpha=0.85)`

- Draws the bubbles (points).
 - `alpha=0.85` adds slight transparency.
 - Makes overlapping bubbles easier to distinguish.
-

5. `geom_text(aes(label=Team))`

- Displays team names (T1, T2, T3, etc.).

- Places labels above each bubble for easy identification.
-

6. `scale_x_log10()`

- Applies a logarithmic scale to the Lines of Code axis.
 - Compresses very large LOC values.
 - Makes both small and large projects easy to compare.
 - Prevents smaller values from appearing crowded together.
-

7. `scale_size(range=c(4,15))`

- Controls the minimum and maximum bubble size.
 - Larger bubbles represent teams with more developers.
-

8. `scale_color_gradient()`

- Applies a color gradient based on software complexity.
 - Lower complexity is shown with lighter colors.
 - Higher complexity is shown with darker colors.
-

9. `labs()`

- Adds the graph title.
 - Adds a subtitle.
 - Labels the X-axis and Y-axis.
 - Displays legend titles.
-

10. `theme_minimal()`

- Applies a clean and professional theme.
 - Removes unnecessary background elements.
 - Improves readability.
-

Aesthetic Mapping

- X-axis → Lines of Code (Log Scale)
 - Y-axis → Number of Bugs
 - Bubble Size → Number of Developers
 - Bubble Color → Software Complexity
 - Labels → Team Names
-

Interpretation

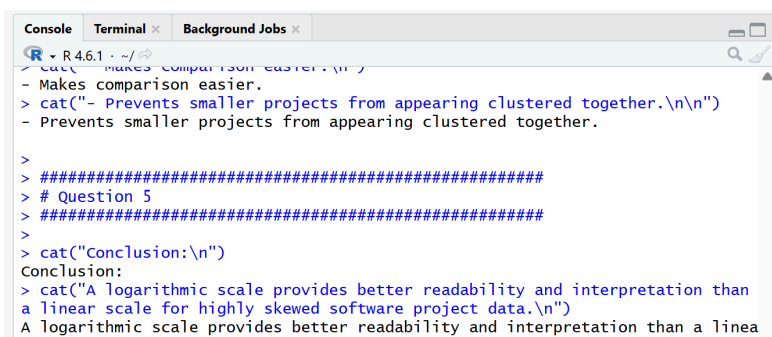
- Projects with more Lines of Code generally have more Bugs.
 - Larger bubbles represent teams with more Developers.
 - Darker-colored bubbles indicate higher software complexity.
 - Team labels help identify individual projects.
-

Why Use a Logarithmic Scale?

- Lines of Code values vary from 12,000 to 1,800,000.
 - A logarithmic scale compresses large values.
 - Improves readability and comparison.
 - Makes trends easier to identify.
-

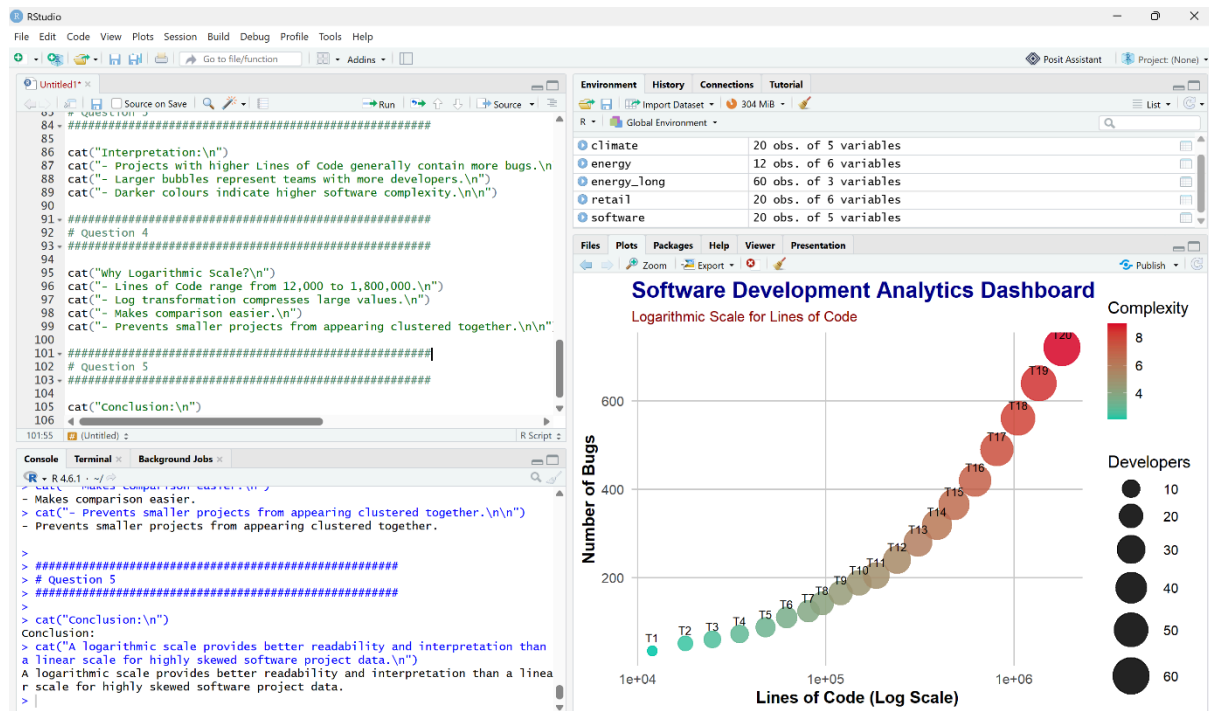
Advantages of the Visualization

- Displays multiple software metrics in one graph.
- Bubble size highlights team size.
- Color represents software complexity.
- Log scale improves comparison across projects.
- Easy to identify trends between code size and bugs.



```
Console | Terminal x | Background Jobs x
R - R 4.6.1 - ~/R
> cat("Makes comparison easier.\n")
- Makes comparison easier.
> cat("- Prevents smaller projects from appearing clustered together.\n\n")
- Prevents smaller projects from appearing clustered together.

>
> #####
> # Question 5
> #####
>
> cat("Conclusion:\n")
Conclusion:
> cat("A logarithmic scale provides better readability and interpretation than
a linear scale for highly skewed software project data.\n")
A logarithmic scale provides better readability and interpretation than a linea
```



QUESTION 5

CODE:

```
# Load Library
```

```
library(ggplot2)
```

```
# Dataset
```

```
companies <- data.frame(
```

```
  Company=paste("C",1:20,sep=""),
```

```
  Sector=c("IT","Banking","Healthcare","Retail","Energy",
```

```
  "IT","Banking","Healthcare","Retail","Energy",
```

```
  "IT","Banking","Healthcare","Retail","Energy",
```

```
  "IT","Banking","Healthcare","Retail","Energy"),
```

```
  Revenue=c(420,580,720,850,980,
```

```
  1150,1320,1480,1650,1820,
```

```
  2050,2280,2550,2840,3150,
```

```
  3480,3860,4250,4680,5200),
```

```
  ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
```

```
  20.4,17.8,14.2,10.5,22.1,
```

```
  24.5,19.8,16.7,13.4,26.2,
```

```
  28.1,21.5,18.3,15.2,30.4),
```

```
  Employees=c(1500,3200,2100,4200,2800,
```

```
  5100,6800,3900,7200,4500,
```

```
  8100,9500,6200,10300,7200,
```

```
  11800,13200,8600,14500,16000),
```

```
  MarketCap=c(8500,12000,9800,7600,14500,
```

```
  18200,21500,16800,13200,24500,
```

```
28500,32500,24800,19200,38200,  
42500,46800,35600,29800,52000)  
)
```

```
#####
```

```
# Integrated Visualization
```

```
#####
```

```
ggplot(companies,
```

```
  aes(x=Revenue,
```

```
      y=ProfitMargin,
```

```
      size=MarketCap,
```

```
      color=Sector))+
```

```
geom_point(alpha=0.85)+
```

```
geom_text(aes(label=Company),
```

```
  vjust=-1,
```

```
  size=3.8,
```

```
  color="black")+
```

```
scale_size(range=c(5,18))+
```

```
scale_color_manual(values=c(
```

```
"IT"="#4361EE",
```

```
"Banking"="#F72585",
```

```
"Healthcare"="#06D6A0",
```

```
"Retail"="#F4A261",
```

```
"Energy"="#FFBE0B"
```

```
))+
```

```
labs(
```

```
  title="Financial Analytics Dashboard",
```

```
  subtitle="Revenue vs Profit Margin with Market Capitalization",
```

```
  x="Revenue (₹ Crores)",
```

```
  y="Profit Margin (%)",
```

```
  size="Market Capitalization",
```

```
  color="Industry Sector"
```

```
)+
```

```
theme_minimal(base_size=15)+
```

```
theme(
```

```
  plot.title=element_text(
```

```
    face="bold",
```

```
    size=20,
```

```
    color="darkblue"),
```

```
  plot.subtitle=element_text(
```

```
    size=13,
```

```
    color="darkred"),
```

```
  axis.title=element_text(face="bold"),
```

```
  legend.position="right",
```

```
  panel.grid.major=element_line(color="grey80"),
```

```
panel.grid.minor=element_blank()
)
```

```
#####
```

```
# Question 2
```

```
#####
```

```
cat("Aesthetic Mapping:\n")
cat("Revenue      -> X-axis\n")
cat("Profit Margin -> Y-axis\n")
cat("Market Cap   -> Bubble Size\n")
cat("Industry Sector-> Colour\n")
cat("Company      -> Labels\n\n")
```

```
#####
```

```
# Question 3
```

```
#####
```

```
cat("Interpretation:\n")
cat("- Companies with higher revenue generally have higher profit margins.\n")
cat("- Larger bubbles indicate higher market capitalization.\n")
cat("- Different colours distinguish industry sectors.\n\n")
```

```
#####
```

```
# Question 4
```

```
#####
```

```
cat("Continuous Variables:\n")
cat("Revenue, Profit Margin, Employees, Market Capitalization\n\n")
```



```
cat("Categorical Variable:\n")
```

```
cat("Industry Sector\n\n")
```

```
#####
```

```
# Question 5
```

```
#####
```

```
cat("Conclusion:\n")
```

```
cat("Using position, size and colour together allows multiple variables to be visualized  
simultaneously with minimal perceptual bias.\n")
```

EXPLANATION:

1. `library(ggplot2)`

- Loads the ggplot2 package.
- Used to create attractive and professional visualizations.

2. `companies <- data.frame(...)`

- Creates a data frame named companies.
- Stores information about:
 - Company
 - Sector
 - Revenue
 - Profit Margin
 - Employees
 - Market Capitalization

3. `ggplot(companies, aes(...))`

- Initializes the graph.
- Maps variables to visual aesthetics.

- Revenue → X-axis.
 - Profit Margin → Y-axis.
 - Market Capitalization → Bubble size.
 - Sector → Bubble color.
-

4. `geom_point(alpha=0.85)`

- Draws the bubbles (points).
 - `alpha=0.85` makes the bubbles slightly transparent.
 - Helps when bubbles overlap.
-

5. `geom_text(aes(label=Company))`

- Displays company names (C1, C2, C3, etc.).
 - Places labels above each bubble for easy identification.
-

6. `scale_size(range=c(5,18))`

- Controls the minimum and maximum bubble size.
 - Larger bubbles represent companies with higher Market Capitalization.
-

7. `scale_color_manual()`

- Assigns different colors to each industry sector.
- Makes sectors easy to distinguish.

Example:

- IT → Blue
 - Banking → Pink
 - Healthcare → Green
 - Retail → Orange
 - Energy → Yellow
-

8. `labs()`

- Adds the graph title.
 - Adds a subtitle.
 - Labels the X-axis and Y-axis.
 - Displays legend titles for Market Capitalization and Industry Sector.
-

9. theme_minimal()

- Applies a clean and professional theme.
 - Removes unnecessary background elements.
 - Improves graph readability.
-

Aesthetic Mapping

- X-axis → Revenue
 - Y-axis → Profit Margin
 - Bubble Size → Market Capitalization
 - Bubble Color → Industry Sector
 - Labels → Company Names
-

Interpretation

- Companies with higher revenue generally have higher profit margins.
 - Larger bubbles represent companies with higher market capitalization.
 - Different colors distinguish different industry sectors.
 - Company labels help identify each company.
-

Variables Used

Continuous Variables

- Revenue
- Profit Margin
- Employees
- Market Capitalization

Categorical Variable

- Industry Sector
-

Advantages of the Visualization

- Displays multiple financial variables in a single chart.
- Bubble size highlights company market value.
- Colors clearly differentiate industry sectors.
- Easy to compare company performance.
- Provides a professional and informative financial dashboard.

